

Differentiating Implicit Relations

Answer Key

AP Calculus AB / BC

Quick Answers

1. $-\frac{x}{\sqrt{25-x^2}}$
2. $-\frac{4}{3}, 0$
3. $-\frac{4}{5}, 0$
4. $\frac{4}{5}, 0$

5. $-\frac{4}{13}, 0$
6. $-\frac{3}{8}, 0$
7. $2, -2, 0$
8. $-\frac{25}{(25-x^2)^{\frac{3}{2}}}$

Curriculum

Level: AP Calculus AB / BC

FUN-3.D–FUN-3.E – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 14 tagged checks.

1. $x^2 + y^2 = 25$ — and the explicit check.

Implicitly. Differentiate both sides with respect to x . The term y^2 carries a factor $\frac{dy}{dx}$ by the chain rule, and the constant differentiates to 0:

$$2x + 2y \frac{dy}{dx} = 0 \quad \implies \quad 2y \frac{dy}{dx} = -2x \quad \implies \quad \frac{dy}{dx} = -\frac{x}{y}.$$

The explicit check. On the upper half, $y = \sqrt{25 - x^2}$, so substituting gives $\frac{dy}{dx} = \frac{-x}{\sqrt{25 - x^2}}$. Differentiating $\sqrt{25 - x^2}$ directly by the chain rule gives $\frac{1}{2}(25 - x^2)^{-1/2}(-2x)$, which is the same expression. The two methods agree, which is the point of the exercise: implicit differentiation is not a different derivative, only a different route to it.

$$\boxed{\frac{dy}{dx} = -\frac{x}{y} = \frac{-x}{\sqrt{25 - x^2}}}$$

2. $xy = 12$ at $(3, 4)$.

On the curve? $3 \cdot 4 = 12$, so $xy - 12 = 0$ at that point ✓. Do this first — a slope formula evaluated at a point that is not on the curve is a number with no meaning.

Differentiating. The left side is a product, so

$$y + x \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{y}{x}.$$

At the point. $\left. \frac{dy}{dx} \right|_{(3,4)} = -\frac{4}{3}$. The hyperbola falls steeply here, which matches a slope of magnitude greater than 1.

$\frac{dy}{dx} = -\frac{y}{x}; \quad \left. \frac{dy}{dx} \right _{(3,4)} = -\frac{4}{3}$

3. $x^2 + xy + y^2 = 7$ at $(1, 2)$.

On the curve? $1 + 2 + 4 = 7$ ✓.

Differentiating. The middle term needs the product rule:

$$2x + \left(y + x \frac{dy}{dx} \right) + 2y \frac{dy}{dx} = 0.$$

Collect the $\frac{dy}{dx}$ terms: $(x + 2y)\frac{dy}{dx} = -(2x + y)$, so $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$.

At the point. $-\frac{2(1) + 2}{1 + 2(2)} = -\frac{4}{5}$.

$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}; \quad \left. \frac{dy}{dx} \right _{(1,2)} = -\frac{4}{5}$

4. Folium: $x^3 + y^3 = 9xy$ at $(2, 4)$.

On the curve? $8 + 64 = 72$ and $9(2)(4) = 72$ ✓.

Differentiating. The right side is a product, so both sides need care:

$$3x^2 + 3y^2 \frac{dy}{dx} = 9y + 9x \frac{dy}{dx}.$$

Collect: $(3y^2 - 9x)\frac{dy}{dx} = 9y - 3x^2$, so

$$\frac{dy}{dx} = \frac{9y - 3x^2}{3y^2 - 9x} = \frac{3y - x^2}{y^2 - 3x}.$$

At the point. $\frac{3(4) - 4}{16 - 6} = \frac{8}{10} = \frac{4}{5}$.

$\frac{dy}{dx} = \frac{3y - x^2}{y^2 - 3x}; \quad \left. \frac{dy}{dx} \right _{(2,4)} = \frac{4}{5}$

5. $x^2y + y^3 = 10$ at $(1, 2)$.

On the curve? $1 \cdot 2 + 8 = 10$ ✓.

Differentiating. x^2y is a product *and* contains y , so it needs the product rule with a chain-rule factor on the second piece; y^3 needs the power and chain rules:

$$\left(2xy + x^2 \frac{dy}{dx}\right) + 3y^2 \frac{dy}{dx} = 0.$$

Collect: $(x^2 + 3y^2) \frac{dy}{dx} = -2xy$, so $\frac{dy}{dx} = -\frac{2xy}{x^2 + 3y^2}$.

At the point. $-\frac{2(1)(2)}{1 + 3(4)} = -\frac{4}{13}$.

$$\boxed{\frac{dy}{dx} = -\frac{2xy}{x^2 + 3y^2}; \quad \left. \frac{dy}{dx} \right|_{(1,2)} = -\frac{4}{13}}$$

6. Ellipse $x^2 + 4y^2 = 25$ at $(3, 2)$ — **tangent line.**

On the curve? $9 + 4(4) = 25$ ✓.

Differentiating. $2x + 8y \frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{2x}{8y} = -\frac{x}{4y}$.

At the point. $-\frac{3}{4(2)} = -\frac{3}{8}$.

Tangent line. Point-slope form through $(3, 2)$ with that slope:

$$\boxed{\left. \frac{dy}{dx} \right|_{(3,2)} = -\frac{3}{8}; \quad y = -\frac{3}{8}(x - 3) + 2}$$

Accept the equivalent slope-intercept form $y = -\frac{3}{8}x + \frac{25}{8}$.

7. Horizontal tangents on $x^2 + xy + y^2 = 12$.

Differentiating. As in problem 3, $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$.

Setting the numerator to zero. A horizontal tangent needs $\frac{dy}{dx} = 0$, so $2x + y = 0$, that is $y = -2x$ — provided the denominator $x + 2y$ is not also zero there.

Back into the relation. Substituting $y = -2x$ into $x^2 + xy + y^2 = 12$ gives

$$x^2 - 2x^2 + 4x^2 = 3x^2 = 12 \implies x = \pm 2.$$

With $y = -2x$ this gives the points $(2, -4)$ and $(-2, 4)$.

Confirming. At $(2, -4)$: $4 - 8 + 16 = 12$ ✓, and the denominator is $2 + 2(-4) = -6 \neq 0$, so the tangent really is horizontal there. The point $(-2, 4)$ checks the same way by symmetry.

$$\boxed{(2, -4) \text{ and } (-2, 4)}$$

8. Challenge — second derivative of $x^2 + y^2 = 25$.

Differentiating $\frac{dy}{dx} = -\frac{x}{y}$ again. By the quotient rule, treating y as a function of x :

$$\frac{d^2y}{dx^2} = -\frac{y \cdot 1 - x \frac{dy}{dx}}{y^2}.$$

Substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{y^2 + x^2}{y^3}.$$

Using the relation. On this curve $x^2 + y^2$ is not just any expression — it equals 25. That substitution is what collapses the answer: $\frac{d^2y}{dx^2} = -\frac{25}{y^3}$.

In terms of x , and the concavity. On the upper semicircle $y = \sqrt{25 - x^2}$, so the second derivative is $-25(25 - x^2)^{-3/2}$. There $y > 0$, so the value is negative everywhere on that arc: the upper semicircle is concave down, which is what the picture shows.

$\frac{d^2y}{dx^2} = -\frac{25}{y^3} = -\frac{25}{(25 - x^2)^{3/2}}$
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Grading: full credit requires substituting $\frac{dy}{dx}$ back in and using $x^2 + y^2 = 25$; a student who stops at $-\frac{y^2 + x^2}{y^3}$ has done the calculus but not the simplification.